# ASSIGNMENT 9

## Name:Tanishk Arjun Kalaskar

## Roll no: 2317054

## Batch: C

In [1]:
```
from google.colab import files
uploaded=files.upload()
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Financial Market News.csv to Financial Market News.csv
```

In [2]:
```
import pandas as pd
import numpy as np
```

In [3]:
```
df = pd.read_csv('Financial Market News.csv', encoding='latin-1')
```

In [4]:
```
df
```

Out[4]:

| | Date | Label | News 1 | News 2 | News 3 | News 4 | News 5 | |
|---|---|---|---|---|---|---|---|---|
| **0** | 01-01-2010 | 0 | McIlroy's men catch cold from Gudjonsson | Obituary: Brian Walsh | Workplace blues leave employers in the red | Classical review: Rattle | Dance review: Merce Cunningham | te p |
| **1** | 02-01-2010 | 0 | Warning from history points to crash | Investors flee to dollar haven | Banks and tobacco in favour | Review: Llama Farmers | War jitters lead to sell-off | You |
| **2** | 03-01-2010 | 0 | Comment: Why Israel's peaceniks feel betrayed | Court deals blow to seizure of drug assets | An ideal target for spooks | World steps between two sides intent on war | What the region's papers say | C |
| **3** | 04-01-2010 | 1 | £750,000-a-goal Weah aims parting shot | Newcastle pay for Fletcher years | Brown sent to the stands for Scotland qualifier | Tourists wary of breaking new ground | Canary Wharf climbs into the FTSE 100 | Re |
| **4** | 05-01-2010 | 1 | Leeds arrive in Turkey to the silence of the fans | One woman's vision offers loan lifeline | Working Lives: How world leaders worked | Working Lives: Tricks of the trade | Working Lives: six-hour days, long lunches and... | Po We |
| **...** | ... | ... | ... | ... | ... | ... | ... | |
| **4096** | 20-03-2021 | 0 | Barclays and RBS shares suspended from trading... | Pope says Church should ask forgiveness from g... | Poland 'shocked' by xenophobic abuse of Poles ... | There will be no second referendum, cabinet ag... | Scotland welcome to join EU, Merkel ally says | di Fri |
| **4097** | 21-03-2021 | 1 | 2,500 Scientists To Australia: If You Want To ... | The personal details of 112,000 French police ... | S&P cuts United Kingdom sovereign credit r... | Huge helium deposit found in Africa | CEO of the South African state broadcaster qui... | Br inv tri |
| **4098** | 22-03-2021 | 1 | Explosion At Airport In Istanbul | Yemeni former president: Terrorism is the offs... | UK must accept freedom of movement to access E... | Devastated: scientists too late to captive bre... | British Labor Party leader Jeremy Corbyn loses... | Sh Fire |
| **4099** | 23-03-2021 | 1 | Jamaica proposes marijuana dispensers for tour... | Stephen Hawking says pollution and 'stupidity'... | Boris Johnson says he will not run for Tory pa... | Six gay men in Ivory Coast were abused and for... | Switzerland denies citizenship to Muslim immig... | Pa sta te |
| **4100** | 24-03-2021 | 1 | A 117-year-old woman in Mexico City finally re... | IMF chief backs Athens as permanent Olympic host | The president of France says if Brexit won, so... | British Man Who Must Give Police 24 Hours' Not... | 100+ Nobel laureates urge Greenpeace to stop o... | Bra n |

4101 rows × 27 columns

In [5]:
```
df.isnull().sum()
```

Out[5]:

| | 0 |
|---|---|
| **Date** | 0 |
| **Label** | 0 |
| **News 1** | 0 |
| **News 2** | 0 |
| **News 3** | 0 |
| **News 4** | 0 |
| **News 5** | 0 |
| **News 6** | 0 |
| **News 7** | 0 |
| **News 8** | 0 |
| **News 9** | 0 |
| **News 10** | 0 |
| **News 11** | 0 |
| **News 12** | 0 |
| **News 13** | 0 |
| **News 14** | 0 |
| **News 15** | 0 |
| **News 16** | 0 |
| **News 17** | 0 |
| **News 18** | 0 |
| **News 19** | 0 |
| **News 20** | 0 |
| **News 21** | 0 |
| **News 22** | 0 |
| **News 23** | 1 |
| **News 24** | 3 |
| **News 25** | 3 |

**dtype:** int64

In [6]:
```
import re
import nltk
```

In [7]:
```
nltk.download('stopwords')
```

```
from nltk.corpus import stopwords
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

In [8]:
```
stop_words = set(stopwords.words('english'))
```

In [9]:
```
def clean_text(text):
    text = re.sub(r"http\S+", "", str(text))
    text = re.sub(r"[^a-zA-Z]", " ", text)
    text = text.lower()
    words = text.split()
    words = [word for word in words if word not in stop_words]
    return " ".join(words)
```

In [11]:
```
print(df.columns)
```

```
Index(['Date', 'Label', 'News 1', 'News 2', 'News 3', 'News 4', 'News 5',
       'News 6', 'News 7', 'News 8', 'News 9', 'News 10', 'News 11', 'News
12',
       'News 13', 'News 14', 'News 15', 'News 16', 'News 17', 'News 18',
       'News 19', 'News 20', 'News 21', 'News 22', 'News 23', 'News 24',
       'News 25'],
      dtype='object')
```

In [12]:
```
# Combine all News columns into one single text column
news_columns = [col for col in df.columns if 'News' in col]

df['text'] = df[news_columns].astype(str).apply(lambda x: ' '.join(x), ax
```

In [13]:
```
df['clean_text'] = df['text'].apply(clean_text)
```

In [14]:
```
from sklearn.feature_extraction.text import TfidfVectorizer
```

In [15]:
```
vectorizer = TfidfVectorizer(max_features=5000)
X = vectorizer.fit_transform(df['clean_text'])
```

In [17]:
```
y = df['Label']
```

In [18]:
```
y
```

Out[18]:

| | Label |
|---|---|
| **0** | 0 |
| **1** | 0 |
| **2** | 0 |
| **3** | 1 |
| **4** | 1 |
| **...** | ... |
| **4096** | 0 |
| **4097** | 1 |
| **4098** | 1 |
| **4099** | 1 |
| **4100** | 1 |

4101 rows × 1 columns

**dtype:** int64

In [20]:
```
from sklearn.model_selection import train_test_split
```

In [21]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
```

In [22]:
```
from sklearn.linear_model import LogisticRegression
```

In [23]:
```
model = LogisticRegression()
model.fit(X_train, y_train)
```

Out[23]:
```
▾ LogisticRegression
LogisticRegression()
```

In [24]:
```
y_pred = model.predict(X_test)
```

In [25]:
```
from sklearn.metrics import accuracy_score, classification_report
```

In [26]:
```
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.48721071863581
              precision    recall  f1-score   support

           0       0.43      0.35      0.38       376
           1       0.52      0.61      0.56       445

    accuracy                           0.49       821
   macro avg       0.47      0.48      0.47       821
weighted avg       0.48      0.49      0.48       821
```

In [28]:
```
import yfinance as yf
```

In [36]:
```
#Get Stock Data (Example: Tesla)
```

In [29]:
```
stock = yf.download("TSLA", start="2023-01-01", end="2023-12-31")
```

```
/tmp/ipython-input-3034771787.py:1: FutureWarning: YF.download() has chang
ed argument auto_adjust default to True
  stock = yf.download("TSLA", start="2023-01-01", end="2023-12-31")
[*********************100%***********************]  1 of 1 completed
```

In [35]:
```
# Calculate Daily Return
```

In [30]:
```
stock['Daily_Return'] = stock['Close'].pct_change()
```

In [32]:
```
import matplotlib.pyplot as plt
```

In [34]:
```
# Plot Stock Price
```

In [33]:
```
plt.figure(figsize=(10,5))
plt.plot(stock['Close'])
plt.title("TSLA Stock Price")
plt.show()
```

In [ ]: